COMP 3311
DATABASE MANAGEMENT SYSTEMS

MIDTERM REVIEW EXERCISES

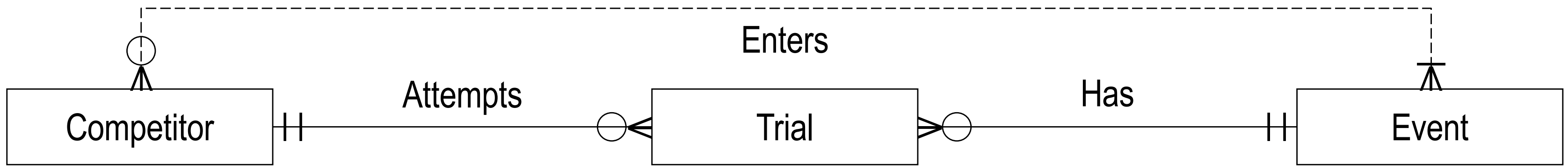
MIDTERM TEST TOPICS

- Database Design
- Relational Database Design
- Relational Algebra (RA)
- Structured Query Language (SQL)

QUESTION 1

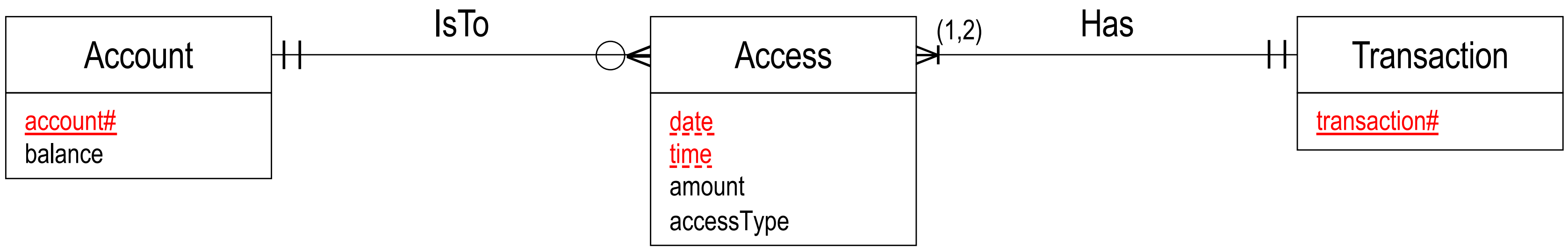
The three entities Competitor, Event and Trial are used to schedule and score athletic competitions such as gymnastics, diving and figure skating. A competitor is described by a unique competitor number and name. An event is described by a unique name. Each trial has a number that is unique for a given competitor and event. An athletic competition can have several events and competitors. Each competitor can enter one or more events. Each event can have several competitors. The focal point of the competitions are the trials. Each trial is an attempt by one competitor to turn in the best performance possible in one event. A competitor receives an overall score for each trial attempted.

Complete the E-R diagram to show how the entity types are related. Show relationship constraints and relationship attributes, if any. The identifying relationship(s) for any weak entities should be clearly indicated.



QUESTION 2

An outline of the reduction of a banking E-R schema to relation schemas is given below. Complete the reduction for each relation schema by adding any required additional attributes, underlining the key and writing the referential integrity constraints including the actions that apply to the relation schema, if any, below it. ***Your reduction must minimize the number of relation schemas.***



Account(account#, balance)

Access(*transaction#, account#, date, time*, amount, access-type)
    foreign key account# **references** Account(account#) **on delete cascade**
    foreign key transaction# **references** Transaction(transaction#) **on delete cascade**

Transaction(transaction#)

QUESTION 3

In the following relations keys are underlined and foreign keys are in italics.

Proposal(<u>pid</u>, *sid*, title, area) // The foreign key sid is **not null** and corresponds to the sid of the submitter who **submitted** the proposal.

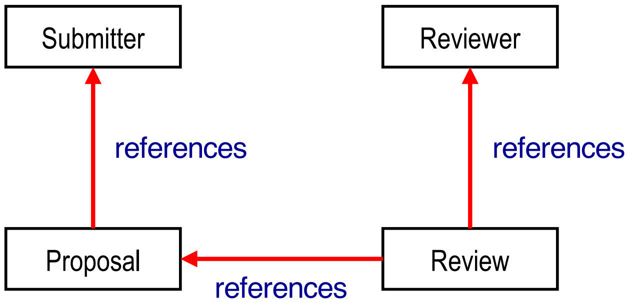
Submitter(<u>sid</u>, name, email) // A submitter may submit several proposals.

Reviewer(<u>rid</u>, name, email, expertise)

Review(<u>*pid, rid*</u>, score) // pid and rid are foreign keys corresponding to the pid of the proposal that was **reviewed** by reviewer rid. The values for score are in the range [1..5]. A reviewer may review several proposals.

QUESTION 3 (CONT'D)

3.1 Given the foreign keys and assuming the referential integrity constraints are included in the SQL **create** statements, what should be the create order?



Create Order

1. Submitter and Reviewer *in any order*
2. Proposal *after Submitter*
3. Review *after Proposal and Reviewer*

Proposal(pid, *sid*, title, area)

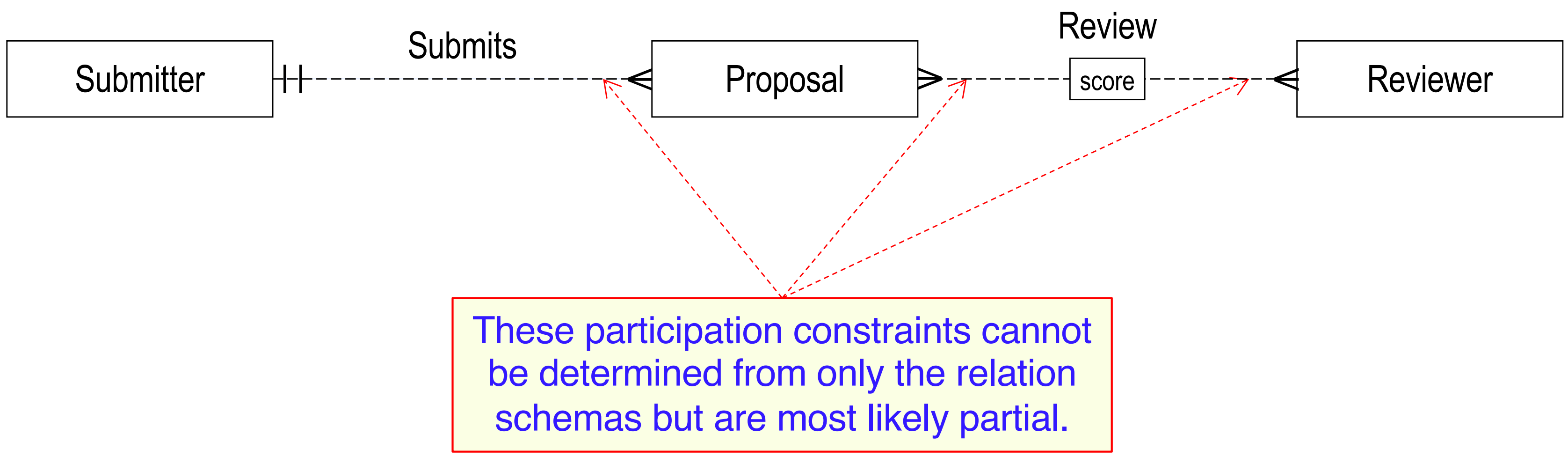
Submitter(sid, name, email)

Reviewer(rid, name, email, expertise)

Review(*pid, rid*, score)

QUESTION 3 (CONT'D)

3.2 Construct an E-R schema that reduces to the given relation schemas. Your E-R schema should minimize the number of entities and relationships required to represent the relation schemas.



Proposal(pid, *sid*, title, area)
Submitter(sid, name, email)
Reviewer(rid, name, email, expertise)
Review(*pid, rid*, score)

QUESTION 4

Given: R(A, B, C, D, E) F = {A→BC, B→AC, AD→E, E→D}

4.1 Which of the following sets is a subset of $\{A, D\}^+$?

a) {A, B}

b) {B, C, D}

c) {E}

d) All the above

e) None of the above

$\{A, D\}^+$	= {A, D}	
	= {A, D, B, C}	using A→BC
	= {A, D, B, C, E}	using AD→E

QUESTION 4 (CONT'D)

Given: R(A, B, C, D, E) $F = \{A \rightarrow BC, B \rightarrow AC, AD \rightarrow E, E \rightarrow D\}$

4.2 Which of the following is a candidate key for R?

a) AD

b) AE

c) BD

d) BE

e) All the above

$AD^+ =$	{A, D, B, C, E}	using $A \rightarrow BC$ and $D \rightarrow E$
$AE^+ =$	{A, E, B, C, D}	using $A \rightarrow BC$ and $E \rightarrow D$
$BD^+ =$	{B, D, A, C, E}	using $B \rightarrow AC$ and $D \rightarrow E$
$BE^+ =$	{B, E, A, C, D}	using $B \rightarrow AC$ and $E \rightarrow D$

QUESTION 4 (CONT'D)

3NF
R is in 3NF *if and only if*
For each FD: $X \rightarrow A$ in F^+:
$A \in X$ (*trivial FD*) ***or***
X is a superkey for R ***or***
A is a prime attribute for R.

Given: R(A, B, C, D, E) $\quad F = \{A \rightarrow BC, B \rightarrow AC, AD \rightarrow E, E \rightarrow D\}$

4.3 For the following decomposition of R, which statement is true?

R_1(A, B, C) $\quad R_2$(A, D, E)

a) The decomposition is 3NF, lossless join and dependency preserving.

b) The decomposition is 3NF, lossless join but not dependency preserving.

c) The decomposition is 3NF, dependency preserving but not lossless join.

d) The decomposition is 3NF, but it is neither lossless join nor dependency preserving.

e) The decomposition is not 3NF and not lossless join, but it is dependency preserving.

R_1 candidate keys: A; B $\quad R_2$ candidate keys: AD; AE

A and B are superkeys in R_1; A, D and E are prime attributes in R_2.

$R_1 \cap R_2 = A$ is a key of R_1. $\{A \rightarrow BC, B \rightarrow AC\}$ in R_1; $\{AD \rightarrow E, E \rightarrow D\}$ in R_2.

BCNF
R is in BCNF *if and only if*
For each FD: $X \rightarrow A$ in F^+:
$A \in X$ (*trivial FD*) ***or***
X is a superkey for R.

QUESTION 4 (CONT'D)

Given: R(A, B, C, D, E) $F = \{A \rightarrow BC, B \rightarrow AC, AD \rightarrow E, E \rightarrow D\}$

4.4 For the following decomposition of R, which statement is true?

$R_1(A, B, C)$ $R_2(A, E)$ $R_3(D, E)$

a) The decomposition is BCNF, lossless join and dependency preserving.
b) The decomposition is BCNF, lossless join but not dependency preserving.
c) The decomposition is BCNF, dependency preserving, but not lossless join.
d) The decomposition is BCNF, but it is neither lossless join nor dependency preserving.
e) The decomposition is not BCNF and not lossless join, but it is dependency preserving.

R_1 candidate keys: A; B R_2 candidate keys: AE R_3 candidate keys: E
A and B are superkeys in R_1; AE is a superkey in R_2; E is a superkey in R_3.
$R_1 \cap R_2 = A$ is a key of R_1. $R_2 \cap R_3 = E$ is a key of R_3. $AD \rightarrow E$ is not preserved.

QUESTION 5

5.1 Construct a single relational algebra expression to find the name of the reviewers who reviewed a proposal in the Database area submitted by Prof. Dimitris (i.e., the submitter name is Dimitris).

$\pi_{\text{Reviewer.name}}((\text{Reviewer JOIN Review}) \text{ JOIN}_{\text{Review.pid=Proposal.pid}}$

$(\sigma_{\text{area='Database'} \wedge \text{name='Dimitris'}} (\text{Proposal JOIN Submitter})))$

Is it necessary to use $\text{JOIN}_{\text{Review.pid=Proposal.pid}}$? Can we just use JOIN?

No. Since Reviewer and Submitter have two attributes in common, name and email, natural join cannot be used to join the two parts of the query.

QUESTION 5 (CONT'D)

5.2 Construct a <u>single relational algebra expression</u> to find the rid of the reviewers who have reviewed proposals only in the area of their expertise (i.e., these reviewers have reviewed at least one proposal and have not reviewed any proposal in an area different from their expertise).

$$\underbrace{\underbrace{(\pi_{rid}\mathrm{Review})}_{\text{rid of all reviewers who reviewed at least one proposal (since they are in Review)}} - \underbrace{(\pi_{rid}(\sigma_{expertise<>area}(\mathrm{Reviewer\ JOIN\ Review\ JOIN\ Proposal})))}_{\text{rid of reviewers who reviewed proposals not in their area of expertise}}}_{\text{rid of reviewers who reviewed proposals only in their area of expertise}}$$

$$\pi_{rid}(\sigma_{expertise=area}(\mathrm{Reviewer\ JOIN\ Review\ JOIN\ Proposal}))$$

Is this a solution?

No. May have reviewed proposals in other areas, not <u>only</u> in area of expertise.

Proposal(<u>pid</u>, *sid*, title, area)

Submitter(<u>sid</u>, name, email)

Reviewer(<u>rid</u>, name, email, expertise)

Review(<u>*pid, rid*</u>, score)

QUESTION 5 (CONT'D)

5.3 Construct a single relational algebra expression that produces the same result as the following SQL query.

```sql
select sid
from Proposal
group by sid
having count(*)>=2;
```

$$\pi_{P1.sid}(\sigma_{\underbrace{P1.sid=P2.sid}_{\text{same sid (i.e., same submitter)}} \wedge \underbrace{P1.pid<>P2.pid}_{\text{different pid (i.e., different proposals)}}}(\rho_{P1}(\text{Proposal}) \text{ X } \rho_{P2}(\text{Proposal})))$$

Proposal(<u>pid</u>, *sid*, title, area)

Submitter(<u>sid</u>, name, email)

Reviewer(<u>rid</u>, name, email, expertise)

Review(*<u>pid, rid</u>*, score)

Use only Oracle SQL constructs presented in the lectures.

QUESTION 5 (CONT'D)

5.4 Construct an equivalent single SQL query *without subqueries* for the following SQL query.

```
select name
from Reviewer
where rid in (select rid
              from Review
              where score=5
                    and pid in (select pid
                                from Proposal
                                where area='Database'));
```

```
select name
from Reviewer R, Review RV, Proposal P
where R.rid=RV.rid
      and RV.pid=P.pid
      and score=5
      and area='Database';
```

```
select name
from Reviewer natural join Review natural join Proposal
where score=5
and area='Database';
```

Proposal(pid, *sid*, title, area)
Submitter(sid, name, email)
Reviewer(rid, name, email, expertise)
Review(*pid, rid*, score)

Use only Oracle SQL constructs presented in the lectures.

QUESTION 5 (CONT'D)

5.5 Construct a single SQL query to find the title and average score of each proposal in the Database area.

```
select title, avg(score)
from Proposal, Review
where Proposal.pid=Review.pid
      and area='Database'
group by Proposal.pid, title;
```

Alternate query

```
select title, avg(score)
from Proposal natural join Review
where area='Database'
group by pid, title;
```

Is it necessary to qualify pid in the group by clause?

Yes. Both Proposal and Review have an attribute named pid.

Does it matter whether we qualify on Proposal or Review?

No. Both Proposal.pid and Review.pid have the same value due to the join condition in the where clause.

Proposal(pid, sid, title, area)
Submitter(sid, name, email)
Reviewer(rid, name, email, expertise)
Review(pid, rid, score)

Use only Oracle SQL constructs presented in the lectures.

QUESTION 5 (CONT'D)

5.6 Construct a single SQL query to find the name, maximum and minimum score of the reviewers who reviewed exactly five proposals.

```
select name, max(score), min(score)
from Reviewer natural join Review
group by rid, name
having count(*)=5;
```

Is it OK to not qualify rid in the **group by** clause?

Yes. Natural join returns only one rid column.

Is it OK to qualify rid in the **group by** clause?

No. There is only one rid column in the query result.

Proposal(pid, *sid*, title, area)
Submitter(sid, name, email)
Reviewer(rid, name, email, expertise)
Review(*pid, rid*, score)

Use only Oracle SQL constructs presented in the lectures.

QUESTION 4 (CONT'D)

4.7 Construct a single SQL query to find, for each proposal, its title, average score and, as a list, the name and score of each reviewer of the proposal. Order the list of reviewers and scores by score ascending and the result by average score descending.

```
select title, avg(score), listagg(name || ' ' || score, '; ') within group (order by score)
from Proposal natural join Review natural join Reviewer
group by pid, title
order by avg(score) desc;
```

Proposal (pid, *sid*, title, area) Submitter (sid, name, email) Reviewer (rid, name, email, expertise) Review (*pid, rid*, score)

QUESTION 5 (CONT'D)

5.7 Express in English the result of the following SQL query.

```
with AverageScores as
(select title, avg(score) as avgScore
 from Proposal natural join Review
 group by pid, title)
select title
from  AverageScores
where avgScore>(select avg(avgScore)
                from AverageScores);
```

Find the title of the proposals whose average score is greater than the average score of all proposals.

Is it OK to use natural join?

Yes. Natural join returns only one pid column.

Proposal(pid, *sid*, title, area)
Submitter(sid, name, email)
Reviewer(rid, name, email, expertise)
Review(*pid, rid*, score)